

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 3: Taxonomic Units: From Kingdom to Species

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	C	B	A	C	D

USE AFTER: PREDICT PHASE — Predict: Kenyan address analogy and Think-Pair-Share

Q1. A Kenyan address goes from the biggest area to the smallest area. Scientists place organisms in groups inside groups in a similar way. Which list of address levels is in the correct order from biggest to smallest?

- A. County, Country, Ward, Sub-county, Village
- B. Country, County, Sub-county, Ward, Village**
- C. Village, Ward, Sub-county, County, Country
- D. Country, Village, Ward, Sub-county, County

Correct answer: B. A country contains counties, which contain sub-counties, then wards, then villages. The address is a nested system, just like taxonomic ranks, which go from the biggest group to the smallest.

USE AFTER: OBSERVE PHASE — Card sorting investigation, Step 1: arrange the rank cards from broadest to most specific

Q2. Which list shows the seven taxonomic ranks in the correct order from the broadest group to the most specific group?

- A. Kingdom, Phylum/Division, Class, Family, Order, Genus, Species
- B. Kingdom, Class, Phylum/Division, Order, Family, Genus, Species
- C. Species, Genus, Family, Order, Class, Phylum/Division, Kingdom
- D. Kingdom, Phylum/Division, Class, Order, Family, Genus, Species**

Correct answer: D. The correct order from broadest to most specific is Kingdom, Phylum/Division, Class, Order, Family, Genus, Species. One distractor reverses Order and Family, one swaps Phylum and Class, and one lists the ranks from most specific to broadest.

USE AFTER: OBSERVE PHASE — Card sorting investigation, Step 3: 'funnel' diagram counting organisms at each rank

Q3. As you move down the ranks from Kingdom to Species, what happens to the number of organisms in each group and to the features they share?

- A. There are fewer organisms in each group and they share fewer features
- B. The number of organisms and the shared features stay the same
- C. There are fewer organisms in each group and they share more features**
- D. There are more organisms in each group and they share more features

Correct answer: C. Each rank is nested inside the one above it, so groups get smaller. Organisms in a smaller group are more alike, so they share more features. The funnel diagram in the activity shows this.

USE AFTER: OBSERVE PHASE — Card sorting investigation, Step 2: match organism-category cards to each rank for the maize plant

Q4. Maize is classified in this order from broadest to most specific: Plantae, Angiospermophyta, Monocotyledonae, Poales, Poaceae, Zea, Zea mays. Which taxonomic rank is Poaceae?

- A. Genus
- B. Family**
- C. Class
- D. Order

Correct answer: B. The ranks run Kingdom (Plantae), Division (Angiospermophyta), Class (Monocotyledonae), Order (Poales), Family (Poaceae), Genus (Zea), Species (Zea mays). Poaceae is the fifth group, so it is the Family.

USE AFTER: EXPLAIN PHASE — Claim-Evidence-Reasoning: How are the taxonomic ranks related from the largest group to the smallest?

Q5. A Kikuyu speaker, a Luo speaker and a Kiswahili speaker each use a different local name for the same animal. Why does the nested ranking system (Kingdom to Species) help them?

- A. It gives every organism one exact place in the system, so all of them can find the same organism whatever its local name**
- B. It replaces all local names so that people stop using them
- C. It puts all animals with similar local names into the same group
- D. It sorts organisms by where they live, so everyone agrees on the habitat

Correct answer: A. Nesting gives each organism one unique position from Kingdom to Species, so people who use different local names can still identify the same organism. The system does not sort by habitat and does not need local names to be abandoned.

USE AFTER: EXPLAIN PHASE — Class discussion after Claim-Evidence-Reasoning: clarifying Phylum and Division

Q6. Animals are placed in a Phylum. What is the name of the rank at the same level, directly below Kingdom, that is used for plants such as maize?

- A. Class
- B. Genus
- C. Division**
- D. Family

Correct answer: C. The second rank is called Phylum for animals and Division for plants. Class, Family and Genus are lower ranks and are not the plant version of Phylum.

USE AFTER: MODEL BUILDING PHASE — Model revision: adding the 'nested ladder' section with the lion (*Panthera leo*) and maize (*Zea mays*) examples

Q7. The lion is classified as Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, Panthera leo. Maize is classified as Plantae, Angiospermophyta, Monocotyledonae, Poales, Poaceae, Zea, Zea mays. Comparing the two lists rank by rank, in how many of the seven ranks do the lion and maize belong to the same group?

- A. 2 ranks
- B. 7 ranks
- C. 1 rank
- D. 0 ranks**

Correct answer: D. The lion and maize already differ at Kingdom (Animalia and Plantae). Because each rank is nested inside the one above, they are in different groups at every rank below it too, so they share no group at any of the seven ranks.